Heaven’s light is our guide

Rajshahi University of Engineering & Technology

Department of Mechatronics Engineering

Course No: MTE 4200
Course Title: Project and Thesis

Defence

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Real-Time Implementation of a Centralized Vision-Based Multi-Robot System for Inter-Robot Collision-Free Navigation and Electromagnetic Object Transportation

OUTLINE

INTRODUCTION

Background

- Industry 4.0 has accelerated demand for intelligent, automated warehouse management solutions.
- Traditional warehouse systems rely on human labor or rigid single-robot automation — both inefficient at scale.
- Multi robotics offers a scalable, fault-tolerant approach where multiple robots cooperate to complete complex logistics tasks.
- Real-time multi-robot coordination combines robotics, control systems, computer vision, and wireless communication.

Why Multi Robotics for Warehouses?

Scalability	Parallelism	Fault Tolerance	Flexibility
Add more robots without redesigning the system	Multiple tasks executed simultaneously	System continues if one robot fails	Adapts to changing warehouse layouts

PROBLEM STATEMENT

Core Problem: How can multiple low-cost microcontroller-based mobile robots coordinate in real-time to execute parallelized warehouse tasks including vision-based localization, collision-free navigation, and item pick-and-drop within a centralized control framework?

Key Challenges Identified

01 Real-Time Coordination

Synchronizing multiple robots without dedicated RTOS — handling latency in wireless communication

02 Vision-Based Localization

Accurately tracking robot positions using an overhead camera with ArUco/color markers in varying light

03 Collision-Free Navigation

Implementing path planning algorithms that avoid inter-robot collisions dynamically

04 Hardware Constraints

Operating reliably within the computational and power limitations of ESP8266 microcontrollers

OBJECTIVES

01 To design and construct two mobile robots with a centralized control architecture.

02 To develop a vision-based localization system using overhead camera tracking and markers.

03 To implement path planning and motion control algorithms for collision-free multi-robot navigation.

04 To integrate item pick-and-drop mechanisms to simulate real warehouse operations

LITERATURE REVIEW

SL No	Year	Methodology	Results	Limitations
1	2025	▪ Reviews AI-driven control strategies for biomimetic robots in locomotion, sensing, and learning. ▪ Highlights use of ML, DL, and RL in biomimetic systems.	▪ AI enhances robot adaptability. ▪ Enables intelligent, flexible behaviors. ▪ Overcomes functional limitations.	▪ Highlights key challenges and future research directions in biomimetic robot control. ▪ Emphasizes RL evolution, adaptive/online control, and neural network-based simulation for optimization.
2	2025	▪ Systematic review using PRISMA on 47 studies from 1248 records across major databases. ▪ Examines DAG-based blockchain integration in smart mobility for EVs, robots, and drone swarms.	▪ DAG improves scalability, latency. ▪ Addresses traditional blockchain limits. ▪ Throughput >1000 TPS, <1s latency.	▪ Challenges: consensus, finality, attacks. ▪ Priorities: benchmarks, hybrid consensus, privacy, microtransaction optimization.
3	2025	▪ Reviews path-planning methods with pros, cons, and applications. ▪ Categorizes algorithms as graphical, sampling-based, or AI-based..	▪ Classified planning methods comprehensively. ▪ Framework for method selection. ▪ Identifies key challenges/prospects. .	▪ Challenges: real-time adaptability and scalability. ▪ Future research: hybrid classical-ML methods, adaptive planning, standardized datasets, HPC use.

[1] H. Jung, S. Park, S. Joe, S. Woo, W. Choi, and W. Bae, "AI-Driven Control Strategies for Biomimetic Robotics: Trends, Challenges, and Future Directions," *Biomimetics*, vol. 10, no. 7, p. 460, 2025.

[2] Y. Bai, S. Lee, and S.-H. Seo, "A Survey on Directed Acyclic Graph-Based Blockchain in Smart Mobility," *Sensors*, vol. 25, no. 4, p. 1108, 2025.

[3] J. Abdouni, T. Jarou, T. Mzili, A. Waga, and K. Bensassi, "Challenges and Constraints in Trajectory Planning for Autonomous Robots," *Iraqi Journal for Computer Science and Mathematics*, vol. 5, no. 1, p. 1274, 2025.

LITERATURE REVIEW(CONT.)

Sl No		Methodology	Results	Limitations
4	2023	▪ Optimizes warehouse layouts for multi-robot systems using traffic flow analysis and scenario generation. ▪ Evaluates efficiency with MAPF simulations and layout diversity comparisons.	▪ Optimized layouts reduce congestion. ▪ Doubles robot capacity/scalability. ▪ Generates diverse, high-throughput layouts.	▪ Limitations: no real-robot validation, static environment assumption. ▪ Focus: layout optimization, not coordinated swarm control.
5	2023	▪ Proposes HI-GWO for robot path planning. ▪ Uses chaos mapping, adaptive updates; tested on benchmarks and real maps.	▪ Superior optimization performance. ▪ Faster, accurate convergence. ▪ Shorter, smoother paths generated. ▪ Robust global search.	▪ Limitations: high computational complexity, sensitive to parameters. ▪ Future: refine algorithm, optimize parameters, test in real-world scenarios.
6	2024	▪ Reviewed case studies and research on swarm robotics in logistics. ▪ Assessed performance, efficiency, and algorithms like PSO, ACO, and Bee.	▪ Enhances operational efficiency. ▪ Speeds order fulfillment. ▪ Improves inventory management.	▪ Challenges: communication, task allocation, safety, AI complexity. ▪ Future: improve reliability, scalability, safety, AI integration, real-world testing.

[4] [4] Y. Zhang, M. C. Fontaine, V. Bhatt, S. Nikolaidis, and J. Li, “Multi-Robot Coordination and Layout Design for Automated Warehousing,” *in Proc. 32nd Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2023, pp. 5503–5511. doi: 10.24963/ijcai.2023/611

[5] L. Liu, L. Li, H. Nian, Y. Lu, H. Zhao, and Y. Chen, "Enhanced Grey Wolf Optimization Algorithm for Mobile Robot Path Planning," *Electronics*, vol. 12, no. 19, p. 4026, 2023.

[6] A. R. Allam, N. R. B. Sridharlakshmi, P. K. Gade, and S. S. M. G. N. Venkata, "Exploring Swarm Robotics for Enhanced Coordination and Efficiency in Logistics Operations," *Robotics Xplore: USA Tech Digest*, vol. 1, no. 1, pp. 137–156, 2024.

LITERATURE REVIEW(CONT.)

Sl No	Year	Methodology	Results	Limitations
7	2023	▪ Developed a UAV-ground robot system for inventory with vision-based localization and uncertainty handling. ▪ Designed a low-cost, modular architecture suitable for warehouse logistics.	▪ Efficient inventory counting. ▪ Single system: 3 mins/2 shelves. ▪ Two systems: 1 min 20s/shelf.	▪ Emulated, simple environment. ▪ No uncertainty/clutter considered. ▪ UAV lacks obstacle avoidance.
8	2025	▪ Reviewed MRTA strategies, focusing on Coalition Formation. ▪ Compared CF algorithms via simulations with 50 robots and 5 tasks.	▪ New MRTA classification. ▪ Proposed CF approach effective. ▪ RL most promising option.	▪ Challenges: long training time, partial observability, non-stationarity. ▪ Future: hybrid methods, deep reinforcement learning, improved coalition strategies.
9	2021	▪ Reviewed early swarm robotics and open challenges. ▪ Examined control methods and criteria for swarm applicability.	▪ Swarm robotics field grew. ▪ Proofs of concept demonstrated. ▪ New research areas identified.	▪ Challenges: limited large-scale real-world tests, lack of reliable engineering methods. ▪ Future: extreme-condition design, ML integration, security, human-swarm interaction.

[7] A. Sales, P. Mira, A. Maria Nascimento, A. Brandão, M. Saska and T. Nascimento, "Heterogeneous Multi-Robot Systems Approach for Warehouse Inventory Management," *2023 International Conference on Unmanned Aircraft Systems (ICUAS)*, Warsaw, Poland, 2023, pp. 389–394, doi: 10.1109/ICUAS57906.2023.10155890.
[8] K. Arjun, D. Parlevliet, H. Wang, and A. Yazdani, "Optimizing Coalition Formation Strategies for Scalable Multi-Robot Task Allocation: A Comprehensive Survey of Methods and Mechanisms," *Robotics (Basel)*, vol. 14, no. 7, p. 93, 2025.
[9] M. Dorigo, G. Theraulaz, and V. Trianni, "Swarm Robotics: Past, Present, and Future," *Proc. IEEE*, vol. 109, no. 7, pp. 1312–1335, 2021.

System Concept

A real-time centralized multi robot system where a master computer coordinates two mobile robots using computer vision for localization, wireless communication (ESP8266 via WebSocket/HTTP), and automated path planning for parallelized warehouse task execution.

Key Ideas

1. Low-cost hardware platform: Built entirely from off-the-shelf components with ESP8266 MCUs
2. Integrated vision pipeline: Real-time robot localization via overhead camera and marker detection
3. Centralized path planner: Collision-free multi-robot navigation in a shared grid environment
4. Electromechanical gripper: Electromagnetic pick-and-drop mechanism integrated with robot chassis
5. End-to-end prototype: Complete working system validated in simulated warehouse environment

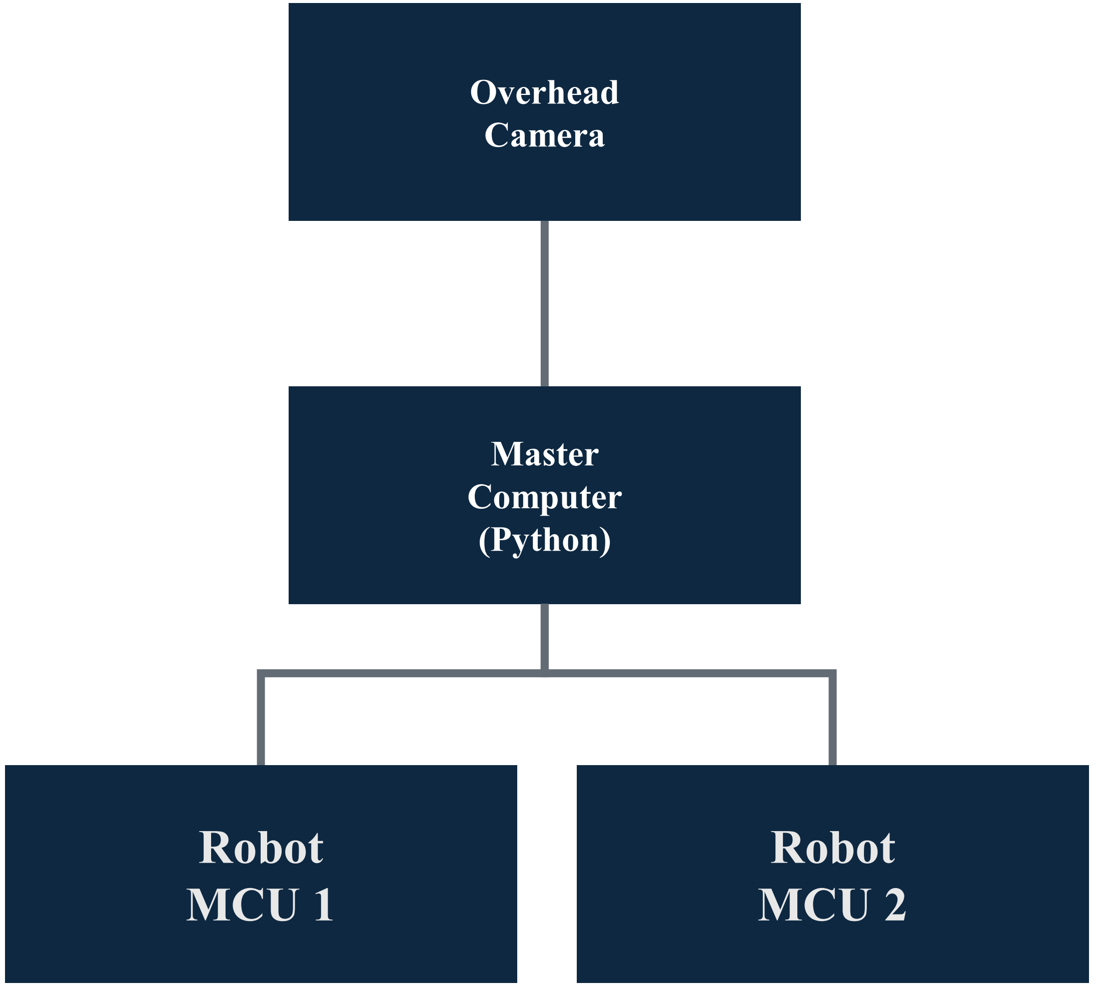


Figure-1: Concept of the system

METHODOLOGY

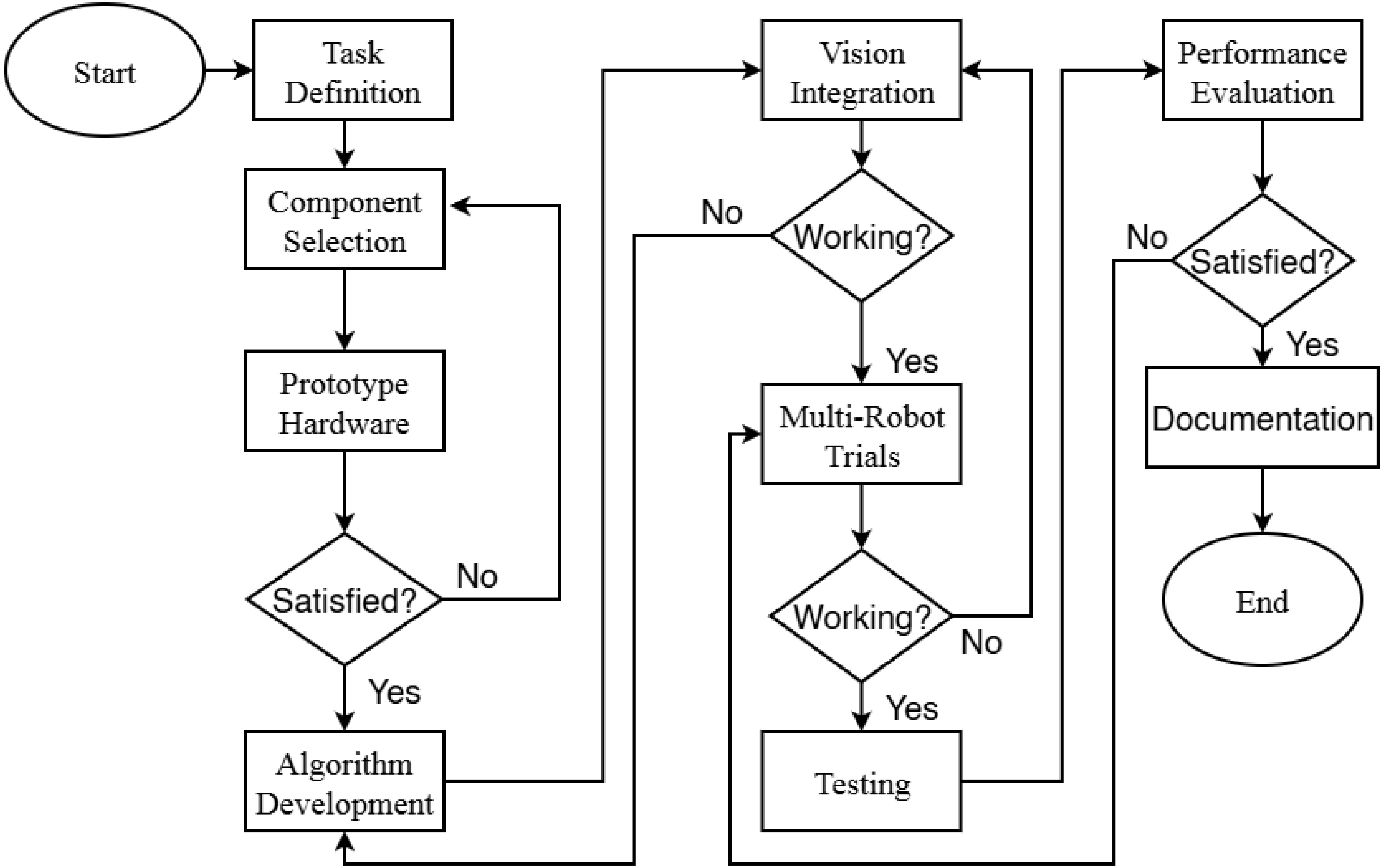


Figure 2: Methodology of the project.

IMPLEMENTATION

Figure-3: Car Design with Internal circuit view, Pallets, Workplace view

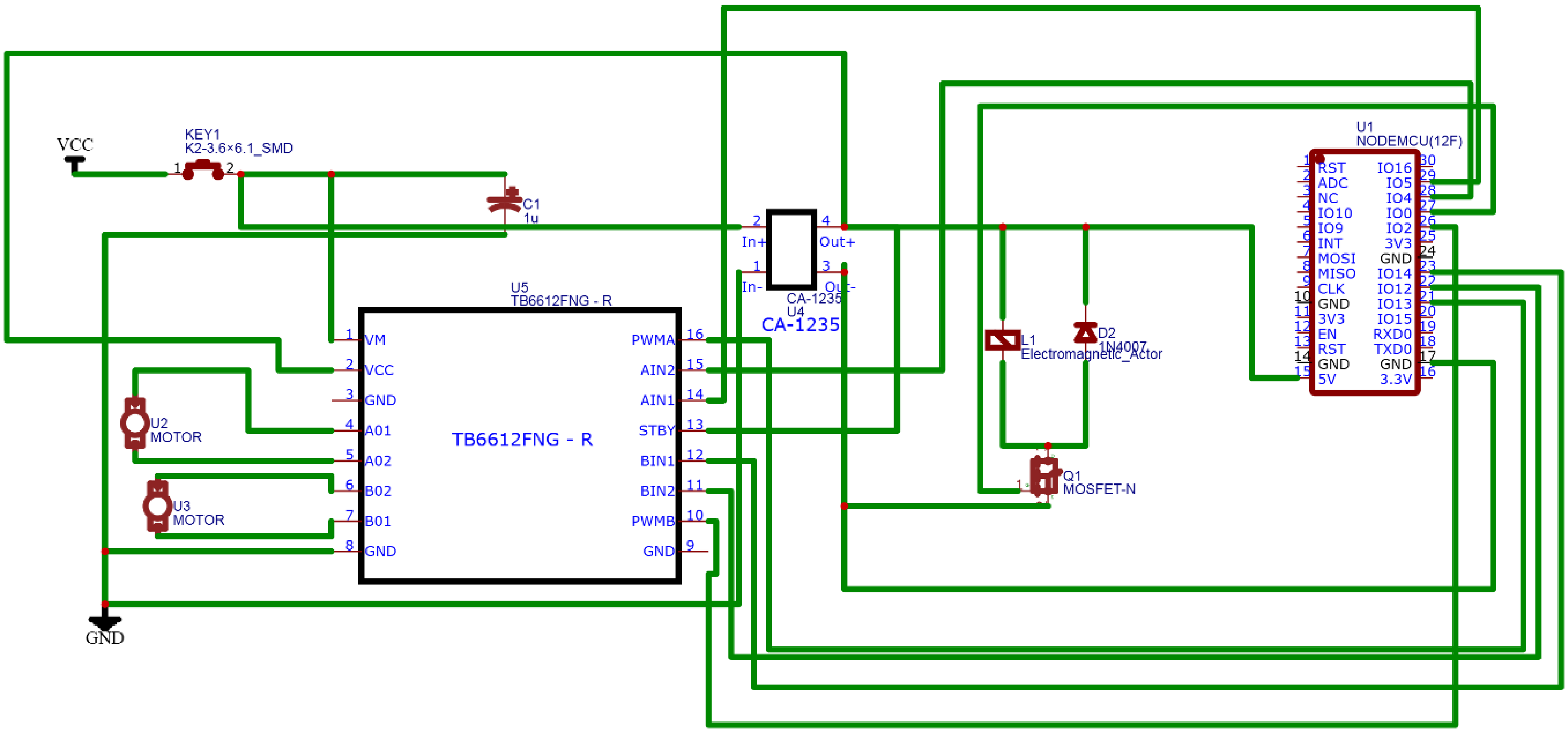
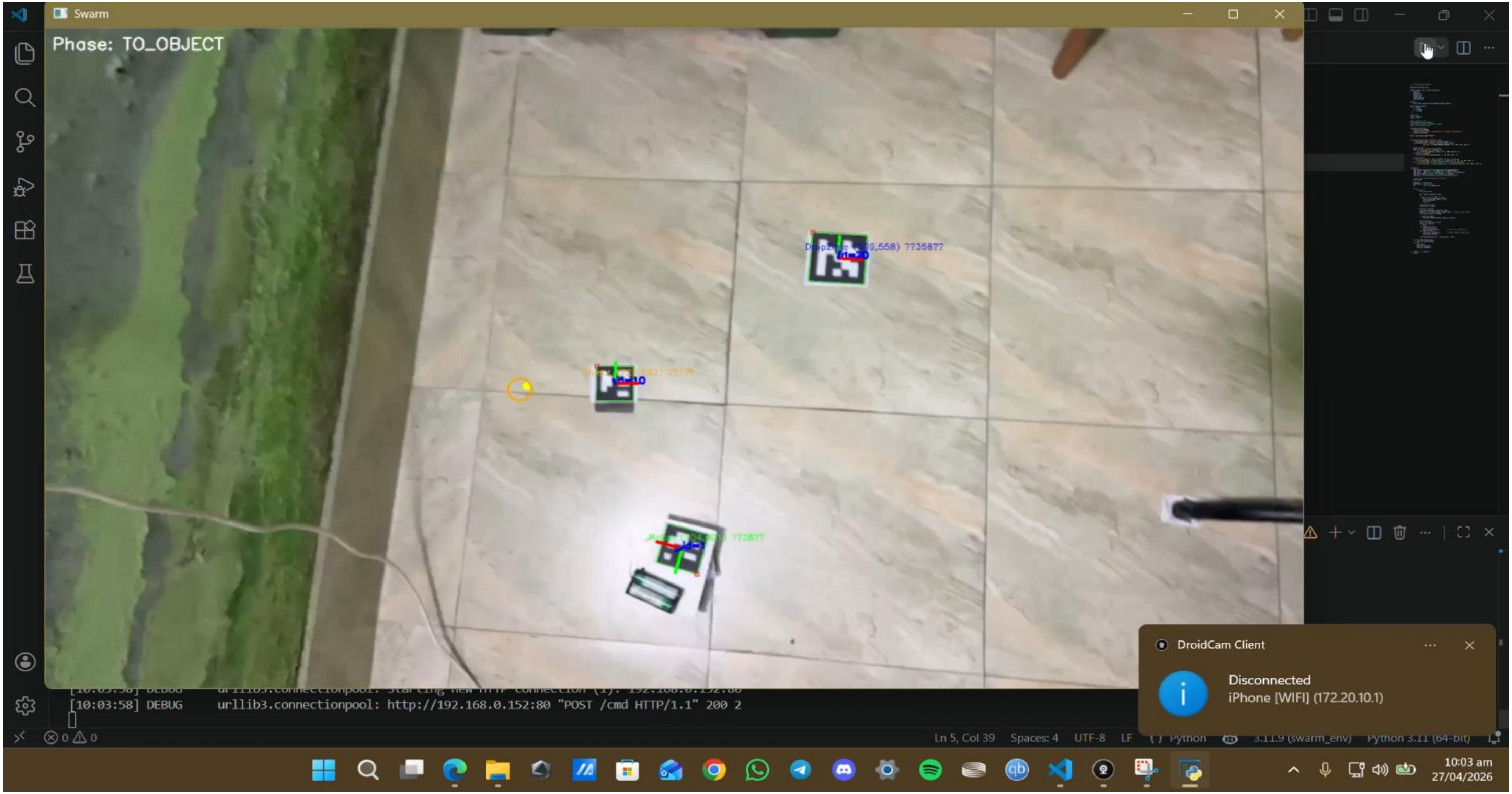
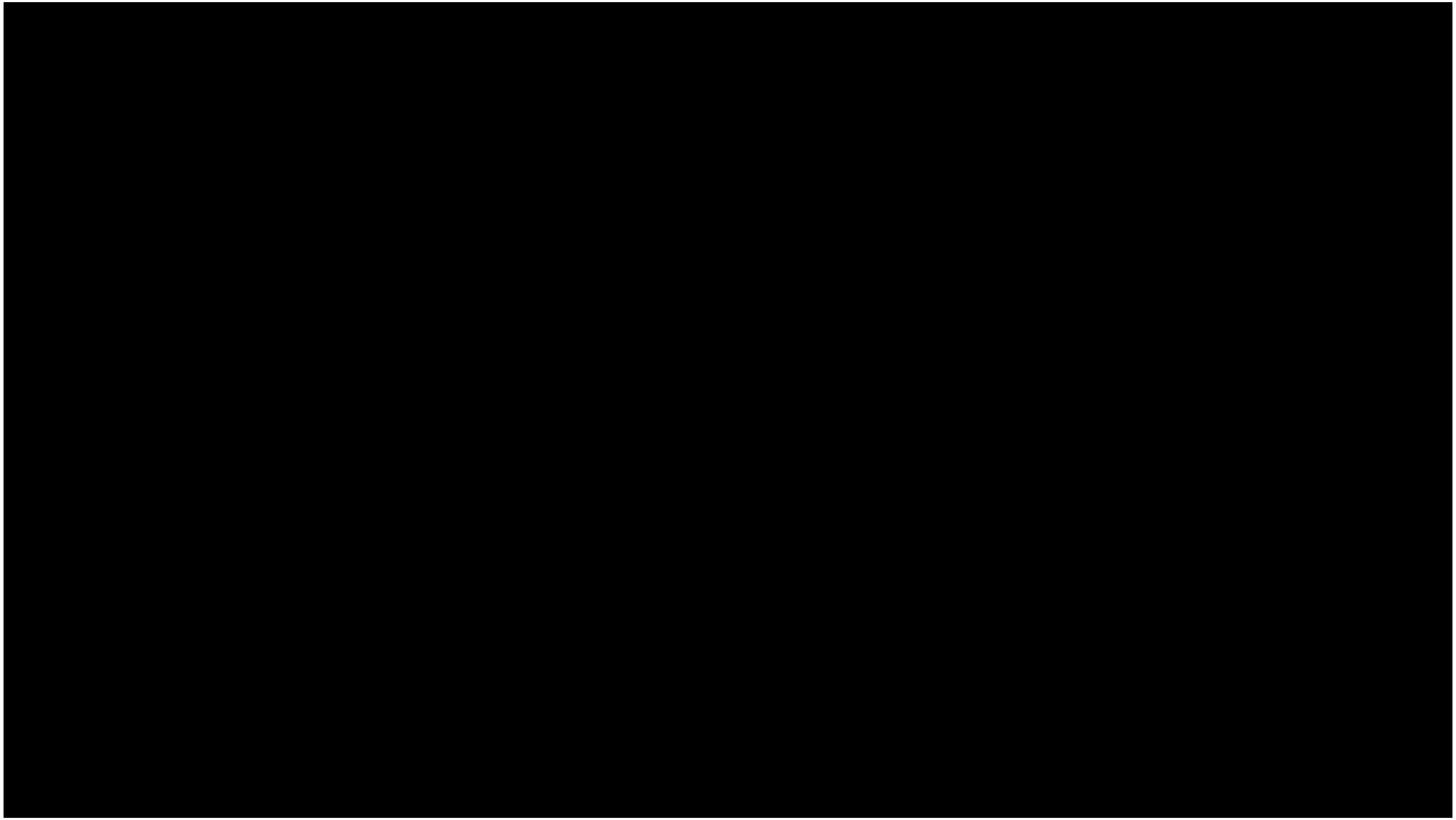


Figure-4: Circuit Connection

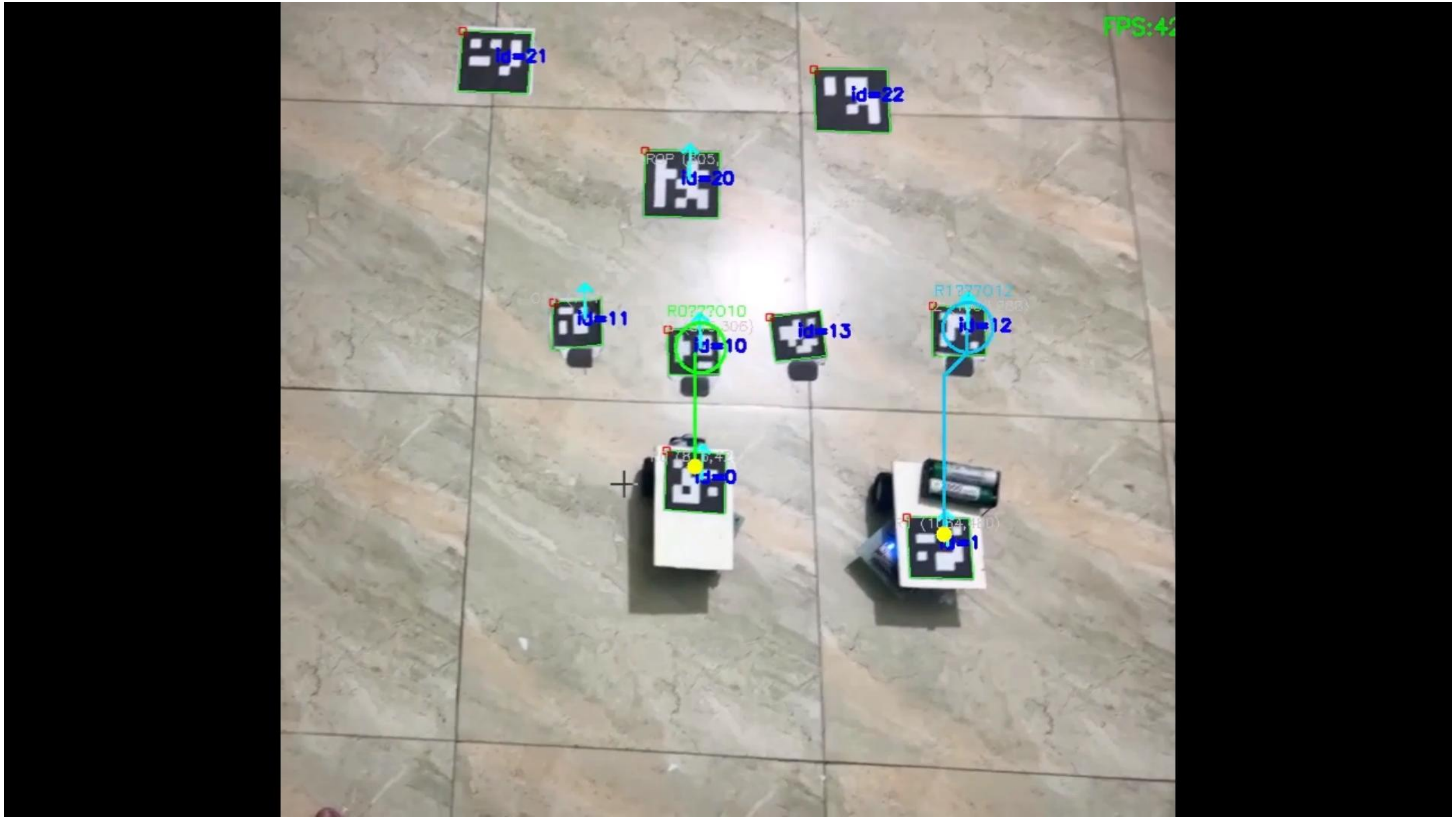
Video-1: Single robot path planning for single objects.



Video-2: Single robot path planning for multiple objects.



Video-3: Two robots working together.



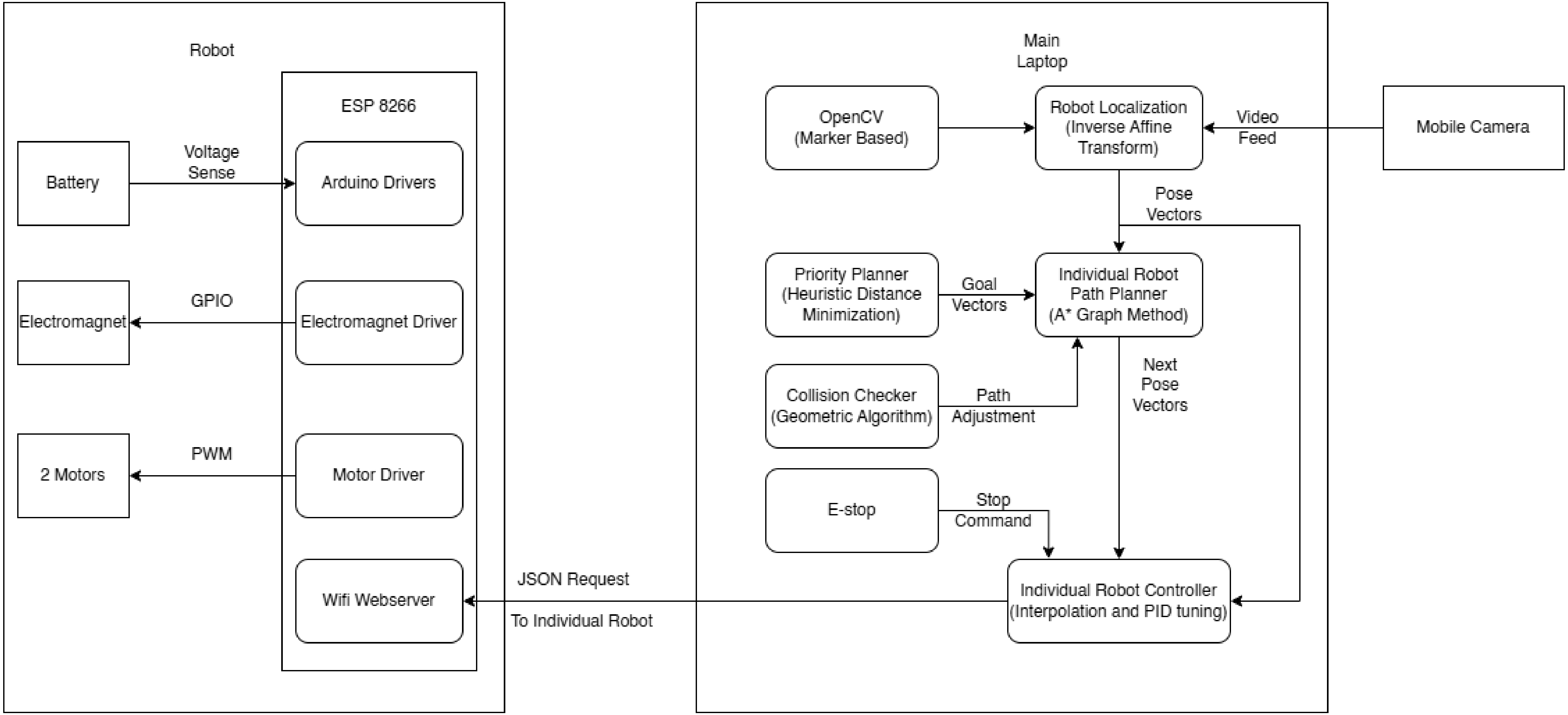


Figure-5: System Architecture

RESULT & ANALYSIS

Table-1: Localization Accuracy

Trial	Error (mm)
1	9
2	11
3	10
Avg	10

Table-2: Task Completion Time

Trial	1 Robot (sec)	2 Robots (sec)
1	136	92
2	140	88
3	132	90
Avg	136	90

Table-3: Performance Summary of A* Path Planning and Controller Execution

Phase	Avg. Waypoints	Controller Behavior	Outcome
Object Targeting	10–18	Smooth velocity scaling	Successful pickup
Drop Zone Return	12–23	High heading adjustment	Successful drop
Recovery/Switch	20+	Frequent re-planning	Adaptive rerouting

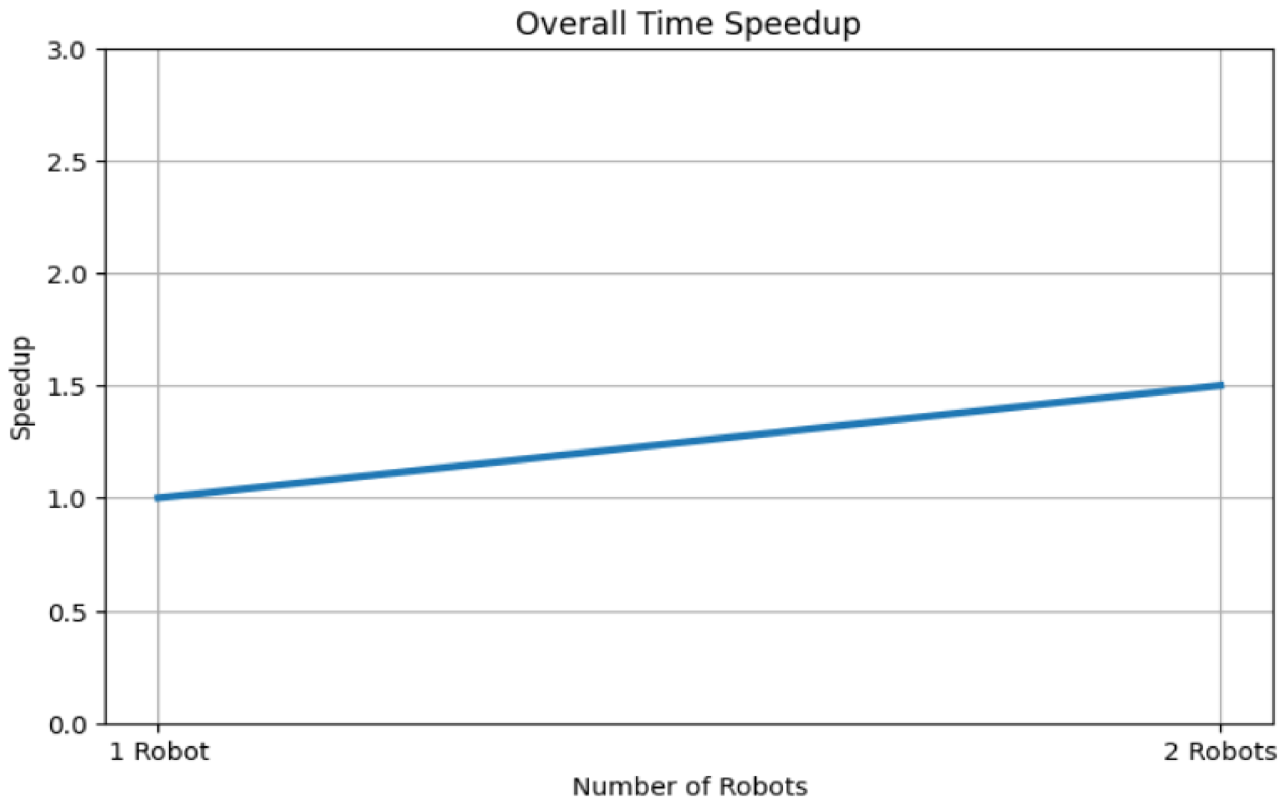


Table-2: Task Completion Time

Trial	1 Robot (sec)	2 Robots (sec)
1	136	92
2	140	88
3	132	90
Avg	136	90

Figure-6: Task Speedup vs Number of Robots

RESULT & ANALYSIS(CONT.)

Table 4: Communication and System Reliability

Parameter	Measured Value
Average Communication Latency (ms)	58
Control Update Frequency (Hz)	20
Successful Command Reception (%)	100
Watchdog Activation During Normal Operation	0
Communication Failure Recovery	Successful

Table 5: Achievement of Research Objectives

Research Objective	**Experimental Outcome**
Design and construction of two mobile robots	Achieved
Vision-based localization using overhead camera and ArUco markers	Achieved
Centralized task allocation, path planning, and collision avoidance	Achieved
Electromagnetic material handling	Achieved with minor mechanical limitation

DISCUSSION

Firmware & Safety

- Non-blocking loop using millis() → always responsive
- Emergency Stop → instant shutdown
- Watchdog (500 ms) → auto-stop
- Fail-safe design.

Hardware & Design

- IRLZ44N MOSFET as switch for electromagnet.
- Flyback diode protection.
- A* = optimal + real-time.
- Scalable for swarm.

Robustness

- ❑ Heartbeat watchdog prevents runaway.
- ❑ E-Stop overrides all.
- ❑ No delay() → always listening.
- ❑ Interrupt motion anytime.

Communication

- ❑ JSON over HTTP → simple & reliable.
- ❑ State variables ensure consistency.
- ❑ Status codes: 200, 400 ,403,413.
- ❑ Robot = command executor.

Performance

- ❑ 9 mm average localization error (99.3% marker detection).
- ❑ A* path planning with real-time replanning (10–23 waypoints).
- ❑ Smooth proportional motion control with stable trajectory tracking.
- ❑ 100% successful task allocation and collision-free coordination.
- ❑ Reliable electromagnet switching with successful object transport.
- ❑ 1.5× faster parallel operation (136 s → 91 s).
- ❑ 58 ms average communication latency with 100% command reception.
- ❑ Stable real-time operation with zero watchdog activations.

Performance

- ❑ A* path planning (10–23 waypoints).
- ❑ Smooth velocity control.
- ❑ Robust Relative localization (~1 cm marker accuracy).
- ❑ Accurate pickup trigger.
- ❑ Scalable Efficiency with 1.5x pickups with two robots.
- ❑ Real time feasibility with 60 ms responsiveness.

CHALLENGES & LIMITATIONS

Wi-Fi Reliability

Medium

ESP8266 operates on 2.4GHz band — susceptible to congestion and occasional packet loss affecting command timing.

Lighting Sensitivity

High

Overhead camera tracking performance degrades under uneven or changing ambient lighting conditions.

Localization Drift

Medium

Absence of wheel encoders causes positional drift over long distances; mitigated via frequent vision correction.

Computational Bottleneck

Low

All vision processing and path planning runs on a single laptop — may bottleneck with increased robot count.

Scale Limitation

High

Current prototype demonstrates 2 robots; requires significant infrastructure upgrade.

Electromagnet Range

Low

Pick mechanism limited to ferromagnetic objects within specific weight/size bounds — restricts item variety.

IMPACT

Research

- ❑ Real-time multi-robot system
- ❑ Practical implementation of swarm robotics

Societal

- ❑ Industrial automation
- ❑ Low-cost scalable solution

Environmental

- ❑ Energy-efficient task execution
- ❑ Reduced resource wastage

CONCLUSION

01 Developed and implemented a centralized vision-based Multi robot system for cooperative parallel material handling.robots.

02 Successfully integrated vision-based localization, task allocation, path planning, motion control, and wireless communication into a unified system.

03 Experimental results demonstrated reliable robot coordination and confirmed the feasibility of parallel task execution using multiple robots.

FUTURE WORK

01 Enhance number of co-robots for studying a large scale co-ordination and fleet management.

02 Use more complex task allocation methods, such as optimization algorithms or machine learning to enhance the efficiency of the system.

03 Enable adaptiveness in obstacle detection and dynamic path replanning for operation in changing environments.

04 Integrate material handling mechanism for various types of objects robotic grippers/hybrid end-effectors.

Thank you.

We are open to Questions.